Task Board - Asana
Project Wiki - GitHub Wiki
Repository - GitHub Repo
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User Guide - Project Archive

CrowdPM Platform: Fullstack PM2.5 Mapping

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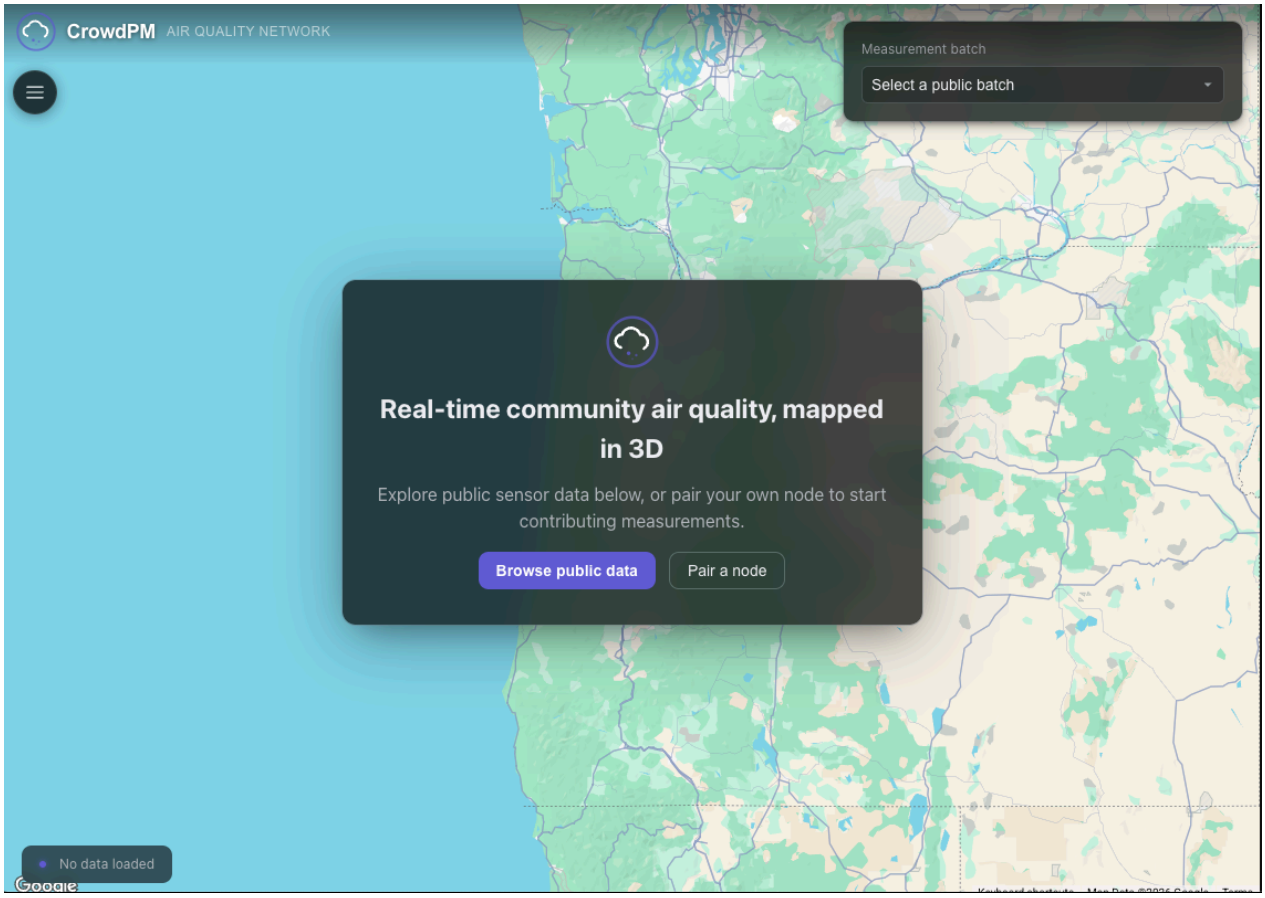
CrowdPM is a community-powered air quality platform. Anyone can visit the map and explore real-time PM2.5 sensor data contributed by people in the community. If you own a CrowdPM sensor node, you can also create an account, pair your device, and start contributing your own measurements.

The Live Map

The map is the first thing you see when you visit the site. It shows PM2.5 air quality measurements from community sensor nodes rendered as a 3D visualization on top of a Google Map.



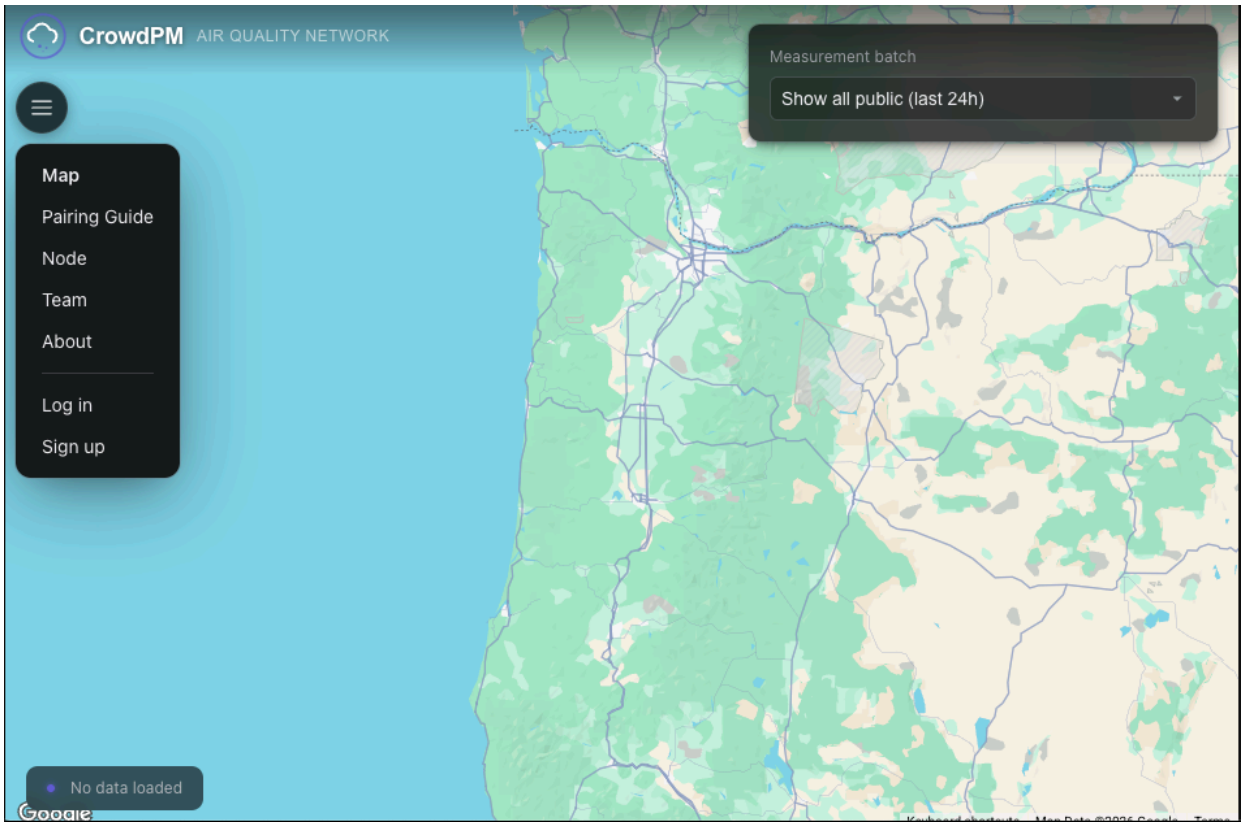
CrowdPM: Live Map View & Main Site Entry Point

Navigation

Below the header on the left side is the navigation menu button (three horizontal lines). Click it to open a dropdown with links to every section of the app:

- Map - Returns to the live 3D air quality map
- Pairing Guide - Step-by-step guide for connecting a sensor node
- Node Hardware - Hardware build and firmware info

- About - Project background and mission
- Dashboard - Your personal device and batch manager (sign-in required)
- Log in / Sign up - Opens the authentication dialog
- Sign out - Signs you out of your account
- Team - Meet the people behind CrowdPM



CrowdPM: Expanded Navigation Bar

Browsing Public Data

You don't need an account to explore public data.

1. Open the **Measurement batch** dropdown in the top-right floating panel.
2. Select **"Show all public (last 24h)"** to load all publicly shared measurements from the past 24 hours across every node.
3. Color-coded points appear on the map. Click any point to jump to that specific batch and measurement.

Selecting a Specific Batch

A batch is a single recording session from one sensor node — a collection of GPS-tagged PM2.5 readings taken during one outing or deployment.

1. Open the measurement batch dropdown
2. Choose any batch from the list. Each entry shows the date/time, the device name, and the number of measurements.
3. The map zooms and tilts to frame the selected batch automatically.

The Timeline Slider

When a specific batch is selected, a Timeline slider appears in the floating panel.

- Drag the slider left or right to step through each measurement in the batch chronologically.
- The currently highlighted point on the map moves to match your slider position.
- Below the slider you can see the exact timestamp, PM2.5 value, GPS coordinates, and precision for the selected point.

The Stats Strip

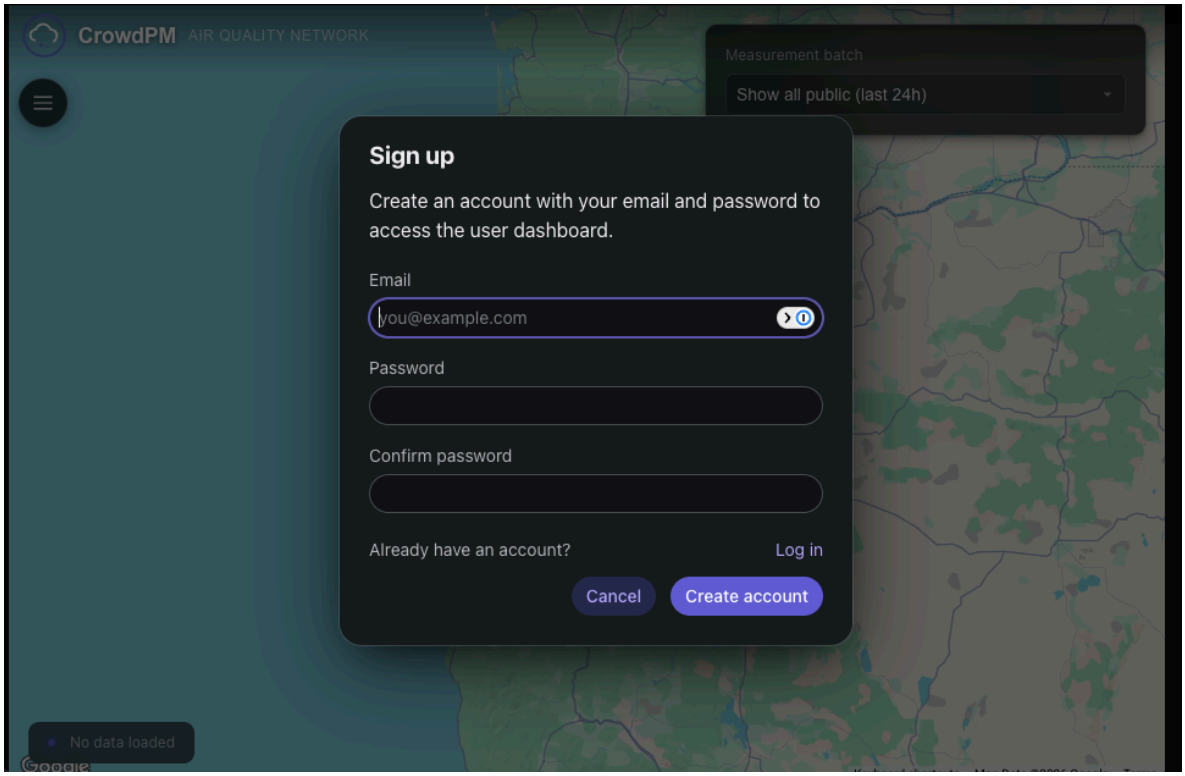
A small strip at the bottom-left of the map shows a summary of what is currently loaded:

- A pulsing dot indicates live data is present.
- The count of measurements (and batches, in "show all" mode) is displayed.

Creating an account

An account is required to pair sensor nodes, manage batches, and access your personal dashboard.

1. Open the hamburger menu (☰).
2. Select Sign up.
3. Enter your email address and choose a password (at least 6 characters).
4. Confirm your password and click **Create account**.



CrowdPM: Signup Form

You are signed in automatically after signing up.

Logging In and Out

To log in:

1. Open the hamburger menu (☰) and select **Log in**.
2. Enter your email and password, then click **Log in.**

To log out:

1. Open the hamburger menu (☰).
2. Select **Sign out**.

Your last-selected batch and cached data are stored per-account in your browser, so your session picks up where you left off when you sign back in.

User Dashboard

The dashboard is your control centre for devices and uploaded data. Sign-in is required.

Overview Cards

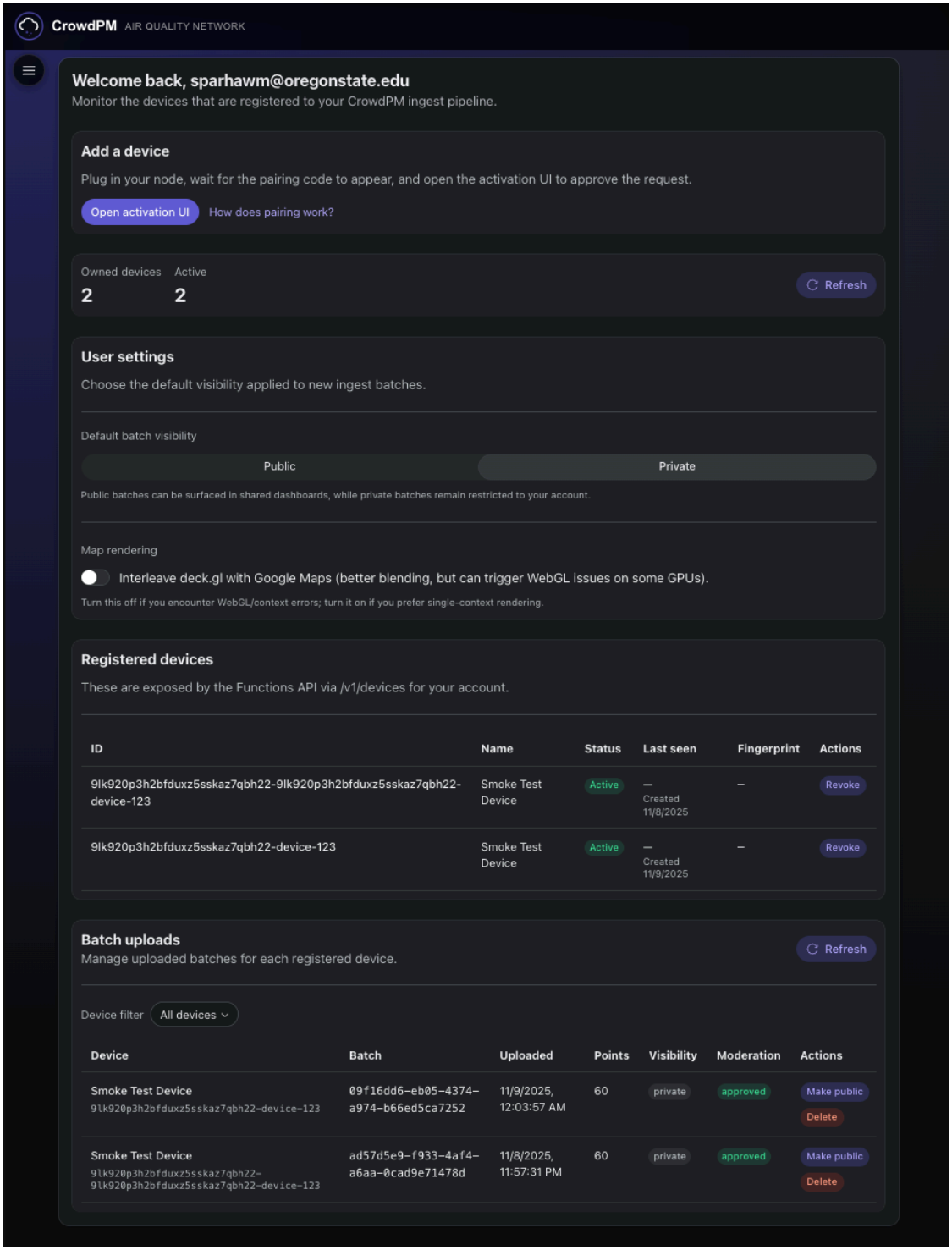
At the top of the dashboard you'll see:

- **Owned devices** — total number of nodes linked to your account.
- **Active** — how many of those nodes currently have a green "Active" status.

Click **Refresh** to pull the latest data from the server.

Use the Open activation UI button to launch the pairing dialog directly from the dashboard. See *Pairing a Sensor Node* for the full walkthrough.

1. **Power on your node** - Plug in the sensor and let it boot. It connects to Wi-Fi and contacts the CrowdPM server automatically.
2. **Read the pairing code** - The node's display (or serial output) shows a short code, e.g. `ABCD-EFGH-J`. This code is valid for 15 minutes.
3. **Open the activation page** - In CrowdPM, open the hamburger menu → Dashboard → Open activation UI, or go directly to `/activate`.
4. **Enter the code** - Type the pairing code from your node into the activation form and click submit. The page shows the device's model, firmware version, and a unique fingerprint for verification.
5. **Approve the device** - Confirm the details match your hardware and click Authorize device. The node detects approval within seconds and begins streaming data.



CrowdPM: Logged-in User Dashboard experience

Pairing a Sensor Node

See *Pairing Guide* on our website for more details.

Pairing connects a physical CrowdPM sensor node to your account. It only needs to be done once per device.

What you'll need

- A CrowdPM-compatible sensor node, powered on and connected to Wi-Fi.
- A CrowdPM account (sign up if you don't have one).
- A web browser on any device (phone, laptop, tablet).

Troubleshooting

Problem	Solution
Code expired	Power-cycle the node to generate a fresh code.
Code not appearing on the node?	Ensure the node has internet access and the display/LED is working.
Session not found? Error	Check for typos — the code uses capital letters and dashes (e.g. `ABCD-EFGH-J`).

Managing Your Batches

The **Batch uploads** section of the Dashboard shows every measurement session uploaded by your devices.

Each row in the table shows:

Column	Meaning
Device	The device name (or ID) that uploaded the batch.

Batch	A unique internal identifier for this recording session.
Uploaded	When the batch was processed by the server.
Points	How many individual PM2.5 measurements are in the batch.
Visibility	**Public** (visible to everyone on the map) or **Private** (only you can see it).
Moderation	Approved (normal) or Quarantined (flagged by an admin for review).

Filtering by Device
Use the Device filter dropdown above the table to show batches from only one specific device.

Changing Batch Visibility
- Click **Make public** to share a batch on the community map.
- Click **Make private** to hide it from public view.

Changes take effect immediately.

Deleting a Batch
Click the red **Delete** button on any row. You will be asked to confirm before the batch and its data are permanently removed.

Video Export

1. Select a batch from the Measurement batch dropdown.
2. Wait for the measurements to load (the timeline slider will appear).
3. Click Render Video in the Video Export card.
4. A progress bar shows the rendering status as the map steps through each measurement point.
5. Once complete, a Download button appears. Click it to save the video as a `.webm` file.
6. Click Re-render to create a new export with the current map state.

7. If you're having trouble with export make sure your "Interleave deck.gl with Google Maps" setting is turned on in the map rendering settings.

Requirements

- The batch must have at least 2 measurements.
- Public batches are exportable when signed out; private batches require you to be signed in as the owner.
- Video export requires a browser that supports canvas recording (Chrome or Edge recommended).

User Settings

Found in the **User settings** card on the Dashboard.

Default Batch Visibility

Choose whether new batches from your nodes default to Public or Private when they are uploaded.

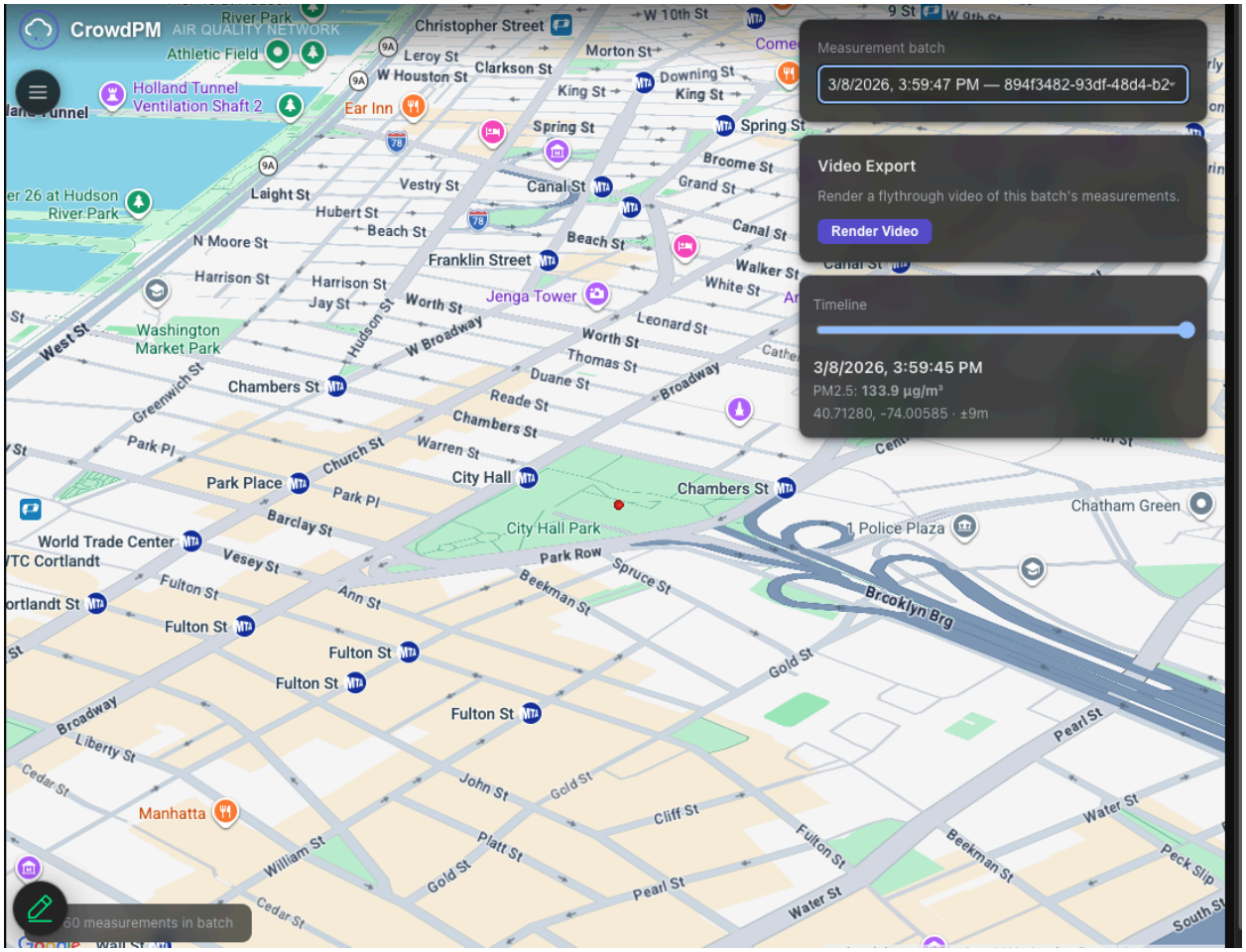
- Public — measurements appear on the community map for everyone.
- Private — measurements are only visible to you on your own dashboard and map view.

You can always change the visibility of individual batches later from the Batch uploads table.

Map Rendering Mode

A toggle for Interleaved map rendering controls how the 3D sensor data layer blends with the Google Maps base layer.

- On (default) — better visual blending between the data layer and the map.
- Off — better compatibility if you encounter graphics glitches or WebGL errors on your device.



CrowdPM: Screenshot of Map and Batch Data